

MUST lecture 2

Multivariate normal distribution

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Multivariate Normal Distribution (MVN)

Vector random variable $X = (X_1, X_2, \dots, X_p)^T$

$$X \sim N_p(\mu, \Sigma)$$

Simultaneous pdf $f_X(x) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right), \quad x \in \mathbb{R}^p$

Mean vector $E[X] = \mu$

Covariance matrix $Cov[X] = \Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} & \dots & \sigma_{1p} \\ \sigma_{12} & \sigma_2^2 & \dots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1p} & \sigma_{2p} & \dots & \sigma_p^2 \end{bmatrix}$

Constant density contours

$$(x - \mu)^T \Sigma^{-1} (x - \mu) = \text{constant} = c^2$$

hyperellipsoid in p dimensions

axis directions e_i eigenvectors for Σ $i=1,2,\dots,p$

axis half-lengths $c\sqrt{\lambda_i}$ eigenvalues for Σ $i=1,2,\dots,p$

Mahalanobis (statistical) distance

$$X \sim N_p(\mu, \Sigma) \Rightarrow$$

$$d^2(x, \mu) \stackrel{\text{def}}{=} (x - \mu)^T \Sigma^{-1} (x - \mu) \sim \chi^2(p)$$

$(1 - \alpha)100\%$ Confidence region for observations

hyperellipsoid

$$d^2(x, \mu) \stackrel{\text{def}}{=} (x - \mu)^T \Sigma^{-1} (x - \mu) \leq \chi^2(p)_\alpha$$

Estimation of MVN parameters

Maximum Likelihood estimate for mean

$$\hat{\mu}_{ML} = \bar{X} = \frac{1}{n} \sum_{j=1}^n X_j \sim N_p(\mu, \Sigma/n)$$

estimate is unbiased and consistent

ML estimate for covariance matrix

$$\hat{\Sigma}_{ML} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^T$$

estimate is not unbiased

$$E[\hat{\Sigma}_{ML}] = \frac{n-1}{n} \Sigma \quad \text{but}$$
$$S \stackrel{\text{def}}{=} \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^T$$

is unbiased and consistent

$$S \sim \frac{1}{n-1} W_p(\Sigma, n-1)$$

Wishart distribution

Model check for MVN

- p = 1** histogram
 boxplot
 QQplot

- p = 2** marginal histograms and boxplots
 scatter plot (look for elliptic shape)
 QQplot for Mahalanobis squared distances against Chi-2(2)

- p > 2** marginal histograms and boxplots
 scatter matrix (look for elliptic shapes)
 QQplot for Mahalanobis squared distances against Chi-2(p)